

Manoel Da Ponte

AI ENGINEER • DATA ENGINEER

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AI and data engineer, co-founder of WiseTwin, where I designed and shipped LLM agents, RAG pipelines and a multitenant SaaS platform. Previously data engineer at TotalEnergies in Denmark and data scientist at CGI. French citizen, based in Switzerland.

Skills

Generative AI & Agents

Data Engineering

Machine Learning

Backend

Frontend

Cloud & DevOps

Real-Time 3D

LLM agents (tool use), RAG, pgvector, LLM-as-judge, SSE, multimodal vision
Idempotent ETL, PostgreSQL, pgvector, SQLAlchemy, Prisma, pandas, Prefect
NLP, computer vision, LLM fine-tuning, deep learning, time series
Python, FastAPI, Node.js, REST APIs, Pydantic, Zod
Next.js, React, TypeScript, Tailwind CSS
Azure (Container Apps, Blob, ACR), Docker, GitHub Actions, Vercel, AWS
PlayCanvas, Gaussian Splatting, GLSL shaders, Unity, Google Maps 3D

Work Experience

WiseTwin

Dunkirk, France (remote)

CO-FOUNDER & LEAD DEVELOPER

March 2024 - Present

- Designed and built a multitenant SaaS platform for industrial safety training, spanning three products (immersive 3D training, document-based courses, prevention tours): 80 000 lines of TypeScript, 124 API routes
- Built an HSE AI copilot: LLM agent (Claude) with 12 tools, agentic loop implemented without a framework, SSE streaming, sensitive actions gated by human confirmation
- Production RAG pipeline: embeddings and pgvector similarity search over 62 726 industrial accidents from the French public ARIA database, with tenant isolation enforced by CI tests
- Automated course generation from PowerPoint: Whisper transcription, multimodal LLM writing, quiz questions verified by a second model (LLM-as-judge)
- Web-based 3D editors in production: Gaussian Splatting (PlayCanvas, GPU selection via custom GLSL shaders) and photorealistic 3D maps (Google Maps 3D, geolocated GLB models)
- Security and compliance: enterprise SSO, TOTP MFA, xAPI/cmi5 and Qualiopi exports; per-organization AI cost governance (token tracking, budgets, quotas)
- Infrastructure: Azure Container Apps, Container Registry and Blob Storage, GitHub Actions CI/CD, Vercel, Neon PostgreSQL
- Technologies: Python, FastAPI, TypeScript, Next.js/React, Claude (Anthropic), OpenAI, pgvector, PostgreSQL, Prisma, Docker, Azure

TotalEnergies

Esbjerg, Denmark

AI ENGINEER & DATA ENGINEER

March 2022 - February 2024

- Delivered end-to-end data projects in small teams (1 to 3 people), covering data management, backend and frontend
- Accident prevention solution: IoT data extraction, NLP analysis of incident reports, centralized data warehouse and LLM fine-tuning
- Invoice control system combining computer vision and predictive analysis for procurement optimization
- Offshore IoT architecture: data warehouse, anomaly detection, Prefect monitoring and automated alerting
- Lightweight tools for large-scale SAP data extraction used by business teams
- Application deployments on Azure and on-premise through CI/CD pipelines
- Technologies: Python, Azure, SQL, NLP, Computer Vision, IoT, Prefect, SAP, React

CGI

Toulouse, France

DATA SCIENTIST

July 2020 - March 2022

- Chatbot for technical documentation access, built on indexing and semantic search (Elasticsearch)
- Cross-system data migration solution on AWS
- Behavioral analytics and personalized recommendations to improve user journeys
- Technologies: Python, AWS, NLP, Elasticsearch, SQL, REST APIs

Education

MSc in Data Science and Engineering

GRADUATE

In-depth data engineering curriculum covering collection, storage and statistical analysis of large-scale data.

Paul Sabatier University, Toulouse III

2019 - 2021

BSc in Applied Mathematics (MIASHS)

GRADUATE

Multidisciplinary curriculum combining applied mathematics, computer science and social sciences.

Jean Jaurès University, Toulouse II

2018 - 2019

Two-year degree (DUT) in Statistics and Business Intelligence

GRADUATE

Statistics and computer science applied to data analysis and decision-support modeling.

University of Caen

2016 - 2018

Selected Projects

WiseAtlas

DESIGNER & DEVELOPER

- Multitenant SaaS editor for interactive territory maps: photorealistic 3D mapping (Google Maps 3D), geolocated 3D models, timeline-based storytelling and rich content blocks
- Production LLM translation feature: batching, rate-limit handling, per-model cost tracking
- Technologies: Next.js, React, TypeScript, Prisma, PostgreSQL, Azure Blob, Claude (Anthropic)

WiseTwin

2025 - 2026

Splat Editor SafetyTour

DESIGNER & DEVELOPER

- Web-based 3D Gaussian Splat editor: cleanup of digitized industrial scenes, GPU selection through GLSL shaders, educational hotspots with quizzes, guided tours and standalone HTML export
- SaaS integration: JWT authentication, Azure storage with signed URLs, scene compression, learner progress analytics
- Technologies: PlayCanvas, WebGL2, GLSL, Next.js, TypeScript, Zustand

WiseTwin

2025 - 2026

Bootstrap-Now

CO-FOUNDER

- SaaS platform supporting new entrepreneurs: business opportunity discovery through multi-channel data aggregation and AI-assisted business plan generation
- Technologies: Next.js, Node.js, PostgreSQL, Prisma, OpenAI API, Vercel

Toulouse, France

2024 - 2025

Languages

French

Native

English

C1 (full professional proficiency, two years in an international environment in Denmark)

Interests

Game development

Personal Unity and C# prototypes, including a multiplayer game

Guitar

Playing since age 12

References

Available upon request.